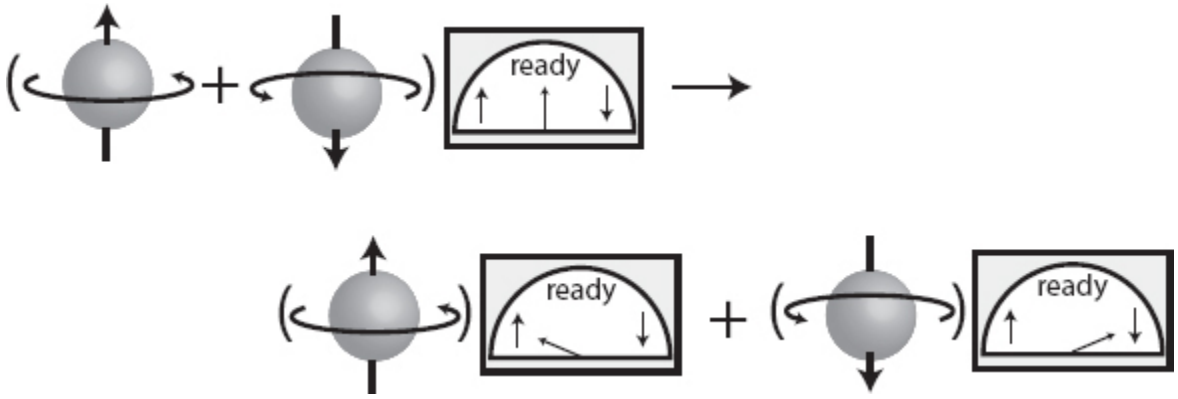
starting from a spin in some superposition and the measurement device in its ready state, we have:



The final state now is an entangled superposition: the spin is up and it was measured to be up, plus the spin is down and it was measured to be down. At this point it's not strictly correct to say "the spin is in a superposition" or "the apparatus is in a superposition." Entanglement prevents us from talking about the wave function of the spin, or that of the apparatus, individually, because what we will observe about one can depend on what we observe about the other. The only thing we can say is "the spin+apparatus system is in a superposition."

This final state is the clear, unambiguous, definitive final wave function for the combined spin+apparatus system, if all we do is evolve it according to the Schrödinger equation. This is the secret to Everettian quantum mechanics. The Schrödinger equation says that an accurate measuring apparatus will evolve into a macroscopic superposition, which we will ultimately interpret as branching into separate worlds. We didn't put the worlds in; they were always there, and the Schrödinger equation inevitably brings them to life. The problem is that we never seem to come across superpositions involving big macroscopic objects in our experience of the world.

The traditional remedy has been to monkey with the fundamental rules of quantum mechanics in one way or another. Some approaches say that the Schrödinger equation isn't always applicable, others say that there are additional variables over and above the wave function. The Copenhagen approach is to disallow the treatment of the measurement apparatus as a quantum system in the first place, and treat wave function collapse as a separate way the quantum state can evolve. One way or another, all of these